LOMAMO FOUNDATION KENYA

CARELINK APP USAGE TRIAL REPORT

Assessment of the digital home-visit platform for caregiver field use

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This report consolidates the field observation/trial findings and the recommendations from the Media & Tech team, including the administrative, data-storage and management questions listed for the system builder in Section 14.

Table of Contents

1. Executive Summary 3
2. Purpose and Scope 4
2.1 Trial Methodology and Participants 4
3. Key Findings and Required Improvements 4
3.1 Patient Not at Home, Absence Reasons and Food-Bag Protocols 4
3.2 Form Logic, Conditional Data Entry and Navigation 6
3.3 Offline Field Operation 7
3.4 Patient-Centred Records and Continuity of Care 7
3.5 Multivitamin Adherence Tracking 8
3.6 Home Visit Frequency and Scheduling Flexibility 9
3.7 Weight Monitoring 9
3.8 Digitisation of All Programme Forms 9
3.9 Caregiver Signatures and Visit Attribution 10
4. Privacy-Preserving Documentation Quality 11
5. Administrative and Data Management Requirements 11
6. Platform Architecture and Administrator Access Model 12
7. Management Dashboards and Weekly Reporting 12
8. Data Verification and Publishing Workflow 13
8.1 Why this workflow is required 13
8.2 The proposed flow, step by step 13
8.3 What this achieves 14
9. Device Management: Samsung Knox (Option Under Exploration) 14
9.1 Capabilities being reviewed 14
9.2 Use cases the exploration will test 15
9.3 Next steps for the evaluation 15
10. Progressive Web App or Standalone Mobile Application? 15
11. Positive Observations 16
12. Prioritized Recommendations 16
13. Tablet and Field-Device Experience 18
14. Questions for the System Builder 18
14.1 Administration, data storage, ownership and security 19
14.2 Verification, signatures, publishing and dashboards 19
14.3 Costs, support and sustainability 19
15. Conclusion 19

1. Executive Summary

LOMAMO Foundation Kenya has trialled the CareLink platform with the intention of improving the way caregivers record and manage home visits. The trial shows that the platform has a strong foundation. It brings many routine home-visit activities into one digital form and, importantly, is intended to support offline field use. Field observations gathered after continued use of the platform refined this assessment beyond navigation and offline behaviour to the detailed clinical and administrative logic that each section of the form should follow. A further set of recommendations from the media team has now been consolidated into this report, so that the system fully reflects the protocol actually followed in the field.

The principal recommendation remains to proceed with CareLink through a structured refinement and caregiver-acceptance phase rather than treating the present version as the final product. The application should be redesigned around the caregiver's actual workflow: selecting the correct patient, recording only information relevant to that visit, saving several forms while offline, reviewing records before synchronisation, and returning to a patient's previous history when needed.

The combined observations point to four main areas of work. First, CareLink needs conditional, or skip, logic within almost every section, so that irrelevant follow-up questions do not appear once an earlier answer has ruled them out. Second, the food-bag protocol must match the reality of the care programme: the reasons a patient is not at home (a hospital visit, unknown, or another reason, and for school-age minors, at school or elsewhere) should be selectable options; the family bag and the couple bag must be clearly defined; every bag issued must be recorded as Bag 1 or Bag 2, since the two bags differ in contents and cost; and issuing a couple bag must immediately capture the spouse's name and whether the spouse was at home to receive it.

Third, records must be patient-centred, with automatic pre-population of recurring details, GPS locations for households so that stand-in caregivers can find them, dates of birth with automatic ages, and a weight measurement for each patient at least once every three weeks or three consecutive visits, tracked by the system. Every form should also carry two signature blocks, one for each of the two caregivers who attend every visit, so that each form shows who actually made the visit (Section 3.9). Fourth, all this data must first pass through LOMAMO Kenya's admin verification and quality control within CareLink before it becomes official: each form a caregiver submits should first reach the administrator's console, where it is scrutinized, just as the admins review and sign the paper forms today, and only a form that has been verified should be published to the central dashboard that other parties relevant to LOMAMO can see (Section 8).

In addition, because the home-visit form changes the manner in which patient data is handled, all the other programme forms must follow it into the system: assessment, hospital in-patient and out-patient, discharge, patient deceased and disclaimer forms. Digitising them is not a matter of convenience but a necessity of the digital transformation agenda, and it is set out in Section 3.8.

A recurring technical fault, in which the Submit button must be tapped several times before a form registers, has also been identified and should be treated as a critical defect. A further priority is the quality of free-text entries such as Patient Condition, Caregiver Extra Notes and Recommendations. Because patient information is confidential, LOMAMO should not rely on external artificial-intelligence services for this purpose. Instead, CareLink should incorporate a privacy-preserving proofreading and terminology function that checks grammar, spelling, punctuation and basic clarity while preserving the caregiver's original meaning. The caregiver should review and approve any suggested correction before it becomes part of the official record.

The trial also surfaced a fundamental platform-architecture issue that must be resolved before rollout: caregivers who log in with a personal email address are currently authenticated as users of the base44 builder platform itself, rather than as end-users of a deployed CareLink application. This is not appropriate for field staff and must be corrected through a proper administrator-and-user access model, described in Section 6.

On device security, the Media & Tech Assistant is exploring Samsung Knox as an option for safeguarding patient data on the field tablets and keeping each device dedicated to CareLink. The exploration is still in progress, and its purpose is to establish whether Knox can secure the deployment efficiently and effectively before any adoption decision is made (Section 9).

Finally, because CareLink completely changes the protocol and procedure for handling patient records at the office, a consolidated list of administrative, data-storage and management questions has been prepared for the builder of the system. These questions are set out in Section 14, and the answers should be received and recorded before rollout decisions are finalised.

2. Purpose and Scope

The trial was undertaken to determine whether CareLink is sufficiently practical for routine home visits and to identify improvements required before implementation. The review considered the home-visit form, offline operation, navigation, patient identification, continuity of records, administrative reporting, data governance, device management and the field tablet experience.

The findings below are based on the LOMAMO CareLink home-visit form as deployed by the builder at the LOMAMO CareLink application site (lomamo-carelink.base44.app), and on a further round of detailed field observations gathered by select caregivers during continued use of the platform. Additional recommendations made by the media team have also been consolidated into this final report.

2.1 Trial Methodology and Participants

The trial was carried out in four of LOMAMO's ten ministry areas: Nyalenda, Ahero, Ombeyi and Manyatta. These four sites were deliberately chosen to reflect the range of conditions caregivers actually work in, comprising two village ministry areas and two slum ministry areas. Three caregivers took part, completing a total of fifteen patient home-visit entries on the CareLink platform using the Samsung A11+ 5G tablet.

Recording a single home-visit entry took an average of five to eight minutes, which the caregivers considered reasonable for a first attempt on the proposed digital system. For a focused usability and workflow pilot of this kind, intended to reveal interface, logic and process issues rather than to produce statistically representative usage data, a sample of three caregivers and fifteen patients across four contrasting ministry areas is an adequate and proportionate basis for the findings in this report. A larger sample, drawn from all ten ministry areas, is nonetheless recommended during the controlled user-acceptance pilot before full rollout, so that the improvements recommended here can be confirmed at full operational scale.

3. Key Findings and Required Improvements

3.1 Patient Not at Home, Absence Reasons and Food-Bag Protocols

When a patient is not found at home during a visit, the form should immediately move the caregiver to the sections that remain relevant, showing only the appropriate fields for food-bag delivery,

multivitamins, visit frequency, hygiene supplies and the caregiver recommendation. This is the protocol followed with the paper forms currently in use.

Reason for absence. Whenever the patient is not at home, the caregiver should select the reason from a short list of tick-able options: hospital visit or admission; unknown; or other, with space to type the specific reason. Because the system will hold each patient's age (Section 3.4), the form should offer additional options whenever the patient is a minor of school age: at school, elsewhere, or hospitalised. When schools are open, children are expected to be at school, and in the worst case a child may have been admitted to hospital due to illness. Repeated absence should remain visible in the patient's history so that programme-policy follow-up can be initiated where applicable.

Food-bag selection and bag types. The caregiver should first select either **Food bag left** or **No food bag left**. Where a bag is left, the caregiver should record the type of bag issued, using the definitions below:

- **Normal (single) bag**, issued to an individual patient.
- **Family bag**, issued for a household parent and child or children (minors) who are all enrolled in the programme.
- **Couple bag**, issued for two adults of the same household who are married and are both enrolled in the programme.
- **Special bag**, issued for the dependants of a hospitalized patient, described further below.

Where a bag is left, the system should also record the recipient and their relationship to the patient, for example family member, relative, son, daughter, husband or wife, grandchild, cousin, nephew, niece, or another trusted household contact, together with the recipient's telephone number where available, or a note that the receiver does not have a phone. This should also accommodate a trusted neighbour who receives the bag at the patient's request when the patient has stepped out briefly and is expected to return the same day.

Bag 1 or Bag 2. Whichever type of bag is issued, whether normal, family, couple or special, the caregiver must indicate whether it was **Bag 1** or **Bag 2**. The two bags have slightly different contents and their contents are bought at different prices, so recording the bag number each time is what allows the office to reconcile what was issued against what was budgeted. This applies to every bag type, including special, family and couple bags, each of which can be issued as either bag.

Special bags for dependants of hospitalized patients. Where “Patient not at home” is selected because the patient has been admitted to hospital, the form should offer a “Does the patient need a special bag?” option, so that dependants left at home, typically children who rely on the patient, can still receive a food bag. LOMAMO does not normally issue food bags to hospital-admitted patients, since such patients are visited through the Foundation's hospital department. The special-bag option exists specifically for their dependants, not for the patient.

Couple-bag issuing flow. When a couple bag is issued, the form should immediately open a spouse field. The caregiver begins typing the spouse's name and the system suggests the enrolled spouse automatically, so it is on record that the patient's spouse also received the shared bag. In the same step, the caregiver should tick whether the other spouse was at home during the visit in which the couple bag was issued. This relies on spouse and couple registration at assessment stage (Section 3.4).

Two related rules apply to couple bags. Where only one spouse is present and the other is not expected to return the same day, for example because they are ill or hospitalised, the caregiver should be able to issue a normal single bag instead, since a couple bag is intended for both parties.

Where one spouse is only briefly absent and expected back the same day, the couple bag may still be issued as usual. That being said, this is subject to change depending on the circumstances the caregiver finds at the patient's home.

No food bag left. An explicit "No food bag left" option should be available for every case in which no bag can be delivered, whether because no one is home, the door is locked, or no trusted recipient is available, so that this outcome is clearly recorded rather than left blank.

Household size. When the caregiver records the number of persons in the household sharing the food bag, the system should compare this figure automatically with the number recorded at the patient's previous visit and indicate whether it represents a change, rather than requiring the caregiver to make this judgement manually. This depends on visit records being stored patient-by-patient, as set out in Section 3.4.

3.2 Form Logic, Conditional Data Entry and Navigation

Beyond general data-entry improvements, the field observations set out precise conditional, or skip, logic that should govern almost every section of the form, so that a caregiver is never asked to complete a field that a previous answer has already ruled out. The required behaviour is summarised below.

SECTION	TRIGGER	REQUIRED BEHAVIOUR
3, Health Risks	"No" selected	Hide the "If yes, please explain" field.
4, Referral Information	"No" selected (not referred)	Hide "Where referred to", "Reasons they were referred" and "Covered by LOMAMO?".
5, Other Drugs	"No" selected	Hide the "If yes, please explain" field.
6, ARV Medications	"No" selected (not on ARVs)	Hide the ARV medication list. When "Yes" is selected, show the structured ARV list described below.
7, TB Medications	"Yes" selected	Show a structured, selectable TB regimen list (Rifampicin, Isoniazid, Pyrazinamide, Ethambutol, Pyridoxine) plus a separate TPT option, instead of free text.
8, Medication Default	"No" selected	Ask "Meds counted?" before "Is the patient in medication default?", since default status can only be judged after counting. Hide "Number of days/pills in default" and "Number of Defaults" when the answer is "No".
9, Wound Care	New first question	Add "Does the patient have a wound?" as the opening question. If "No", hide dressing, saline, wound-picture and related fields entirely.

Structured ARV medication list. For Section 6, once "Yes" is selected, the ARV medications should likewise be offered as a structured, selectable list rather than free text. The list should contain: Abacavir (ABC), Lamivudine (3TC), Dolutegravir (DTG), Tenofovir Disoproxil Fumarate (TDF), Tenofovir Alafenamide (TAF), Zidovudine (AZT, also written ZDV), Emtricitabine (FTC) and Nevirapine (NVP). An **Other** option should be provided so the caregiver can type in a medicine that is not on the list. A structured list speeds up entry, avoids spelling errors in medicine names, and makes reporting on regimens possible. The administrator should be able to update the list as new medicines enter use.

General navigation and data-entry improvements identified earlier in the trial remain valid alongside this conditional logic:

- Provide "Not Applicable" options where a question does not apply, including medication default, multivitamins and fields related to Waterguard.
- Allow blanket and mattress sizes, such as single or double, to be recorded where issued.
- Present "Same" and "Change Frequency" as distinct Home Visit Frequency options.
- Correct the date-picker problem identified during testing, where the caregiver had to tap the following date to record the intended date.
- Introduce an "Incomplete Form" warning and prevent accidental submission or synchronisation until required fields are addressed.

Testers also reported that the Submit button sometimes needs to be tapped several times before a completed form registers as submitted. This appears to be a technical defect rather than a design issue, but it carries real operational risk. An unresponsive submit control can lead to a visit record being lost, delayed or accidentally duplicated. It should be investigated and resolved as a critical, pre-rollout item.

3.3 Offline Field Operation

Offline capability is essential to CareLink's usefulness in the field. The trial found that a form saved offline could not readily be reviewed before synchronisation, and that completing another form required repeated refreshing and reloading. In addition, the "Saved offline" and "Form Cleared" notifications could obstruct the Submit button.

- Allow caregivers to save multiple visit forms offline in a queue.
- Allow every saved offline form to be reopened, reviewed and edited before synchronisation.
- Allow caregivers to continue directly to a new patient form without repeated refreshing.
- Move temporary notifications away from primary buttons, or make them disappear automatically.
- Show clear synchronisation status: saved offline, synchronising, and successfully uploaded.
- Indicate when a synchronised form has been received into the administrator's verification queue, and later whether it has been verified and published or returned for correction (Section 8).
- Retain (as already is) a visible record of forms awaiting synchronisation so that no visit is inadvertently lost.

3.4 Patient-Centred Records and Continuity of Care

The platform should reduce repeated data entry by allowing caregivers to log in, select their ministry area and see the patients served within that area. Caregiver and ministry-area coordinator information should populate automatically. A caregiver should then select the patient and open a blank visit record linked to that patient's existing profile, with previous weekly visits retrievable from the ministry-area database.

The field observations add several specific, practical refinements to this patient-centred approach:

- **Auto-population of recurring fields.** Once a caregiver selects a patient, the new form should automatically carry forward the last visit date, contact information and caregiver name from the patient's existing record, without requiring these to be re-entered. Where any of this information has genuinely changed, the caregiver should be able to update it. The signature is deliberately not carried forward: every visit is signed afresh by the two visiting caregivers, as described in Section 3.9.

- **Relief-caregiver provision.** Where another caregiver stands in for the ministry area's usual caregiver, the form should allow that stand-in caregiver to be recorded, so the visit is correctly attributed.
- **GPS household locations.** At the first visit, and whenever a household relocates, the system should capture and store the GPS coordinates of the patient's home. Household locations should be available to authorised staff so that a caregiver standing in for the ministry-area coordinator can find and visit the household when the coordinator is unavailable or attending to other urgent organisational matters.
- **Dates of birth and ages on record.** Every patient profile should store the date of birth, with the age calculated and displayed automatically. This underpins the absence options for minors described in Section 3.1 (at school, elsewhere or hospitalised) and gives management an accurate picture of the ages of patients in the programme.
- **Caregiver login and access.** Caregivers should log in on the ministry-area tablet using their work email address, select their ministry area, and enter a PIN to access that area's patients. The Director and administrators should retain full visibility across all ministry areas for oversight and management purposes. The caregiver's organisation name and staff role should already be populated in their profile.
- **Spouse and couple registration.** The system should record, from assessment stage, whether a patient's spouse is also enrolled in the programme. This underpins the couple-bag logic in Section 3.1, including the spouse confirmation step when a couple bag is issued.
- **Days-in-programme counter.** Once a patient's enrolment date has been entered, the system should automatically calculate and display the number of days the patient has been in the programme.
- **Discharge summary.** At discharge, the system should automatically compile a summary of everything recorded for the patient from intake through to discharge, drawn from the linked home-visit history, rather than requiring this to be reconstructed manually.
- **Consolidated per-patient dashboard.** Once a caregiver selects a patient, all of that patient's previous forms and key dates should be tabulated and summarised in one simple dashboard. This includes, for example, the date a hygiene bag was last delivered, which can be inferred from the corresponding food-bag record already in the system, and the weight trend described in Section 3.7. The dashboard should also show the verification status (by the admin) of each submitted form, as set out in Section 8. The caregiver should not have to search separate historical forms individually.

Each patient should have a longitudinal digital record from assessment and intake through home visits, hospital visits, discharge or the latest programme activity. This would improve continuity of care, reduce duplication and make PDCA preparation more efficient. The same patient-centred structure should eventually accommodate assessment, home-visit, hospital-visit, discharge, deceased and disclaimer forms, together with relevant supporting documents, as described in Section 3.8.

3.5 Multivitamin Adherence Tracking

The multivitamins section should reflect how multivitamin supplies are actually issued and used in the field, rather than assuming they are only relevant on the day of a visit. Where multivitamins are given on the day of the visit, the existing questions apply as normal. Where they are not given that day, “Taking multivitamins” can still reasonably apply, since a supply may have been issued at an earlier visit, and a date can still be entered for when the last supply was given.

Multivitamin issues are typically intended to last seven days. The system should therefore calculate seven days forward from the date multivitamins were last given, and use this to inform whether the patient would still reasonably be expected to be taking them at the current visit. The caregiver should also be able to indicate that a patient still has multivitamins remaining even after the seven-day period has elapsed, for example where doses were missed. "Is the patient taking multivitamins" and "are they taking them properly, consistently and as intended" should be captured as two distinct questions, since a patient may be taking multivitamins irregularly without this being obvious from a single yes-or-no answer.

3.6 Home Visit Frequency and Scheduling Flexibility

The current Home Visit Frequency field assumes a single, fixed interval per patient, which does not reflect clinical reality. Some patients, particularly those receiving active wound care, may require more than one visit per week: for instance, a dedicated medical visit for wound dressing on one day, and a separate regular home visit, including food-bag delivery and, where needed, a further wound check, on another day. Other patients may require visits only once every two weeks.

The field should therefore allow flexible, patient-specific scheduling rather than a single fixed interval, and the system should offer a smart, needs-based suggestion for frequency. For example, it could prompt twice- or thrice-weekly visits where active wound care is indicated, or fortnightly visits where clinically appropriate. The caregiver or ministry-area coordinator can then confirm or adjust the suggestion based on the patient's actual condition.

3.7 Weight Monitoring

Patient weight is a core indicator of health and nutrition status in the programme, and it must be measured on a regular, verifiable schedule. The required schedule is a weight measurement for each patient at least once every three weeks, or across every three home visits. Stated simply, any three consecutive visits must include at least one weight measurement.

CareLink should therefore record the date of each patient's last weight measurement and automatically prompt the caregiver when the next one is due. The system must also take note of patients whose last three visits passed without a weight measurement being taken, and flag them so that the gap is corrected at the very next visit rather than discovered later. Weight readings should be stored on the patient's record and shown as a simple trend in the per-patient dashboard (Section 3.4), making it easy to see whether a patient is gaining, holding or losing weight over time.

3.8 Digitisation of All Programme Forms

Taking the home-visit form digital changes the manner in which patient data is handled across the whole office. It follows that the other forms used by the programme cannot remain on paper, because that would split each patient's history between two worlds and recreate the very double handling CareLink is meant to remove. Digitising the remaining forms is therefore not a matter of making work easier; it is a necessity of the digital transformation agenda.

The forms concerned are:

- **Assessment form**, which opens the patient's record at intake and captures the baseline information, including their children, spouse and couple registration.
- **Hospital in-patient and out-patient forms**, used when patients are visited through the Foundation's hospital department.
- **Discharge form**, which closes the patient's programme record and triggers the automatic discharge summary described in Section 3.4.

- **Patient deceased form**, which records a death and its date with the dignity and accuracy the case deserves, and updates programme numbers accordingly.
- **Disclaimer forms** and any other consent or acknowledgement documents the office relies on.

Each of these forms should live inside the same patient-centred structure, carry the same conditional logic and pass through the same verification workflow as the home-visit form (Section 8), so that nothing reaches the official record, the central dashboard (final report), without passing through the administrator's review. A practical order of digitisation is recommended: the assessment form first, since it opens every patient record; then the discharge and patient deceased forms, which close records; then the hospital in-patient and out-patient forms; and finally the disclaimer and remaining consent documents. During transition, the paper versions remain the reference copy until each digital form has been tested and signed off.

3.9 Caregiver Signatures and Visit Attribution

On the paper forms, caregivers sign each completed form, and two caregivers attend every home visit. The signature is what shows that the named caregiver actually carried out the visit, and this must not be lost in the move to digital records. CareLink should therefore provide a signing block at the foot of every form, and the digital signature should carry the same weight the paper signature carries today.

Two signature blocks on every form. Each form should close with two signature areas: one for the **lead caregiver**, who owns and submits the form, and one for the **accompanying caregiver** who attended the same visit. The lead caregiver's name populates from her login. The accompanying caregiver is chosen from the caregiver list, with the same type-and-suggest behaviour used for the spouse field in Section 3.1, so the second signer is recorded accurately even when a relief caregiver stands in.

How the signature is captured. Each caregiver signs with a finger or stylus directly on the tablet screen, in her own signature box, at the point of care. The signature is stored as an image attached to that single visit record, together with the date and time it was drawn. It is not a saved image that is stamped onto future forms: every visit is signed afresh, which is what makes the signature meaningful.

Sign-and-submit rule. The form should not submit until both signature blocks are completed. Where a visit was genuinely made by one caregiver alone, the form should offer a solo-visit indication with a brief reason, and this exception is displayed to the administrator at verification. An unexplained missing signature should be flagged, never silently accepted.

Attribution beyond the signature. A drawn signature on its own proves little, because it could be added by anyone holding the tablet. The system therefore binds each signature to a wider set of evidence automatically: the account that was logged in, the ministry area, the tablet used, the GPS household location captured for the visit (Section 3.4), and the exact time of signing. Together, the two signatures and this evidence give LOMAMO stronger proof of who made each visit than the paper form ever could.

Offline behaviour and downstream use. Signatures are captured as part of the form, work offline, and travel with the queued form until synchronisation. Both signatures appear in the administrator's verification console, where the admin confirms they are present just as they check the paper signatures today (Section 8), and they remain visible in the patient's visit history, the discharge summary and any printed or exported copy of the form.

4. Privacy-Preserving Documentation Quality

CareLink should improve the quality of caregiver-written notes without changing what the caregiver actually observed. For free-text fields such as Patient Condition, Caregiver Extra Notes and Recommendations, the preferred solution is a built-in proofreading and terminology checker that operates within LOMAMO's approved application or data environment, rather than transmitting patient notes to an external artificial-intelligence service.

The function should correct or flag spelling, grammar, punctuation and basic clarity issues, while preserving names, dates, medication names, quantities and the substance of the caregiver's observations. It should not create symptoms, diagnoses, treatments or events that were not entered by the caregiver. Suggested changes should be presented for caregiver review, with options such as Accept, Edit or Keep Original.

LOMAMO-specific terminology should be included in the application's approved dictionary so that commonly used programme and medical terms are recognised correctly. This approach would improve professionalism and consistency in documentation while maintaining control over sensitive patient information. It should be presented as a documentation-quality aid, not as clinical decision support.

A further safeguard could be a final documentation check before submission. The system could flag obvious missing information, contradictions or blank fields without rewriting the caregiver's record. This would complement the recommended “Incomplete Form” warning and strengthen overall data quality.

5. Administrative and Data Management Requirements

The introduction of CareLink should eliminate, rather than simply relocate, the duplication created by the present process of completing physical forms and subsequently entering information into spreadsheets. This raises essential questions about where digital data is stored, who controls it and who is authorised to access it. The following should be considered:

- Define data ownership, storage location, retention, access rights and security responsibilities before full rollout.
- Implement the caregiver login sequence described in Section 3.4 (work email, ministry-area selection, PIN) with role-based access, so caregivers see only their own ministry area's patients while the Director and administrators retain full oversight.
- Adopt the two-stage verification model described in Section 8, under which a record reaches the central dashboard only after the administrator has reviewed and published it.
- Maintain an audit trail showing who created or amended a record, and when.
- Provide secure attachment of warning letters, medical receipts and relevant medical documents to the appropriate patient record.
- Ensure that the system can produce reports without requiring administrators to manually reconstruct figures from separate spreadsheets.

Because CareLink completely changes the protocol and procedure for handling patient records at the office, these concerns have been drawn up as a specific list of questions to be put to the builder of the system. The full list is set out in Section 14 and should be answered before rollout decisions are finalised.

6. Platform Architecture and Administrator Access Model

During the trial, it was observed that opening the CareLink web app at lomamo-carelink.base44.app and logging in with a personal email address does not sign the user into CareLink as an end-user at all. Instead, it signs that person into base44 itself, the underlying platform on which CareLink was built, and prompts them, by name, to (for example) build a new app of their own. This is a builder-level login, intended for developers configuring applications, not a caregiver-level login intended for staff using a finished tool in the field. It is not appropriate for caregivers to interact with base44 at this level, and it should not be part of the finished CareLink experience.

This points to a structural gap rather than a cosmetic one. As currently deployed, CareLink does not yet have its own independent, published sign-in layer that is separate from the base44 builder account used to create it. Before rollout, LOMAMO's technical contact or base44 support should confirm that the application can be published with its own authentication system, so that end-users never see or touch the base44 building environment at all.

The recommended access model, consistent with the login sequence already set out in Sections 3.4 and 5, is as follows:

- A single master administrator account, held by the Director or a designated Chief Administrator, controls the CareLink system at the top level, including the base44 (or eventual hosting) account used to build and publish it.
- Within the deployed CareLink application itself, and not within base44, this administrator creates and manages individual caregiver usernames and PINs or passwords for each ministry area.
- Caregivers log into the deployed CareLink application, never into base44, using the username and PIN or password issued to them, and reach only their own ministry area's patients.
- Ministry-area access is controlled by a common PIN known to, and issued by, the administrator, so that access to each area is deliberate and traceable rather than open to anyone with a personal email address.
- The Director and other authorised administrators retain full visibility across all ministry areas for oversight and management, consistent with the role-based access already recommended in Section 5.

This model brings clear order and accountability to system access: one person is ultimately responsible for the platform, every caregiver's access is provisioned deliberately rather than self-created, and no caregiver is ever placed in a position where they could inadvertently alter, duplicate or interfere with the underlying application while trying to record a home visit.

7. Management Dashboards and Weekly Reporting

A central dashboard should be treated as a core CareLink requirement, operating at three levels: an organisation-wide view for the Kenya office, a ministry-area view for coordinators, and the per-patient view described in Section 3.4 for individual caregiver use. The Kenya office should be able to view consolidated information across all ministry areas and drill down into any one of them. Weekly planning indicators should include patients in programme, discharges, children rescued, deceased patients, orphaned children, food bags, patients saved in Christ, patients not found at home, inpatient and outpatient activity, mattresses and blankets, Bibles, hygiene packages and mosquito nets.

Food-bag reporting on the dashboard should show bags by type (normal, family, couple and special) and by bag number (Bag 1 or Bag 2), so the office can reconcile issues against budgets, and weight measurements taken should also be visible so that the monitoring schedule in Section 3.7 can be overseen.

The dashboard should show patient numbers by ministry area and categorise ministry areas into villages and slums, with totals for each category. This would provide management with a consistent source of information for weekly planning, accountability and reporting, rather than relying on manual calculations.

Importantly, the central dashboard should present verified, published data only, as set out in Section 8. Everyone who views it, from caregivers to Board members and the finance and data colleagues, then works from the same approved figures rather than from drafts still awaiting the administrator's review.

8. Data Verification and Publishing Workflow

8.1 Why this workflow is required

Under the current paper arrangement, the administrator goes through every individual home-visit form, scrutinises it and signs against it. Only after that review does the admin enter the data manually into Excel spreadsheets, and it is from those spreadsheets that the data and finance colleagues are able to see and work with the figures. This review-and-sign step is LOMAMO's own quality control, and it must not be lost when the paper forms are replaced. The application should therefore reproduce it digitally, beginning with the caregiver's submission and ending with publication to one central dashboard that everyone relies on.

8.2 The proposed flow, step by step

1. The caregiver completes and submits the digital form. Before submission, both visiting caregivers sign the form in their signature blocks (Section 3.9). The completed form is saved to the patient's record with the status **Submitted, awaiting verification**. The caregiver can see this status, and nothing further is required of them for that visit.

2. The form arrives in the administrator's console. Every submitted form appears in a verification queue that only the administrator can work in. The console lists submissions by ministry area, patient, caregiver and date, so the admin can review them in batches, much as they work through the paper forms today.

3. The administrator scrutinises the data. The admin checks each form against the standards used in LOMAMO's current review: that both caregiver signatures are present, completeness, consistency with the patient's history, correct bag types and numbers, and the quality of the written notes. If something is unclear or wrong, the admin returns the form to the caregiver with a short note, and the status changes to **Returned for correction**. The caregiver sees the note and the reason at the next login or synchronisation, corrects the form and submits it again.

4. The administrator verifies and approves. When the admin is satisfied, they approve the form. The admin's name and the time of approval are recorded on the record, together with the audit trail. This approval is the digital equivalent of their signature against the paper form.

5. The verified record is published to the central dashboard. Approval publishes the record automatically to the central dashboard, which becomes the single source of verified data for the whole organisation. The status shown to the caregiver becomes **Verified and published**.

6. Everyone works from the published data. Caregivers, the Director and administrators, Board members, the finance and audit colleagues and the data team, including the colleagues overseas, can all view the published dashboard according to their roles. Their access is view-only: they see the same verified figures, and no one outside the verification workflow can alter them.

Corrections after publication follow the same path in reverse: the administrator reopens the record, the correction is made, and the record is re-verified and republished, so the dashboard always matches the verified record and the audit trail always shows what changed.

8.3 What this achieves

- The office keeps its proven quality control: nothing becomes official until the administrator has reviewed it, exactly as with the paper forms they sign today.
- Manual entry into Excel spreadsheets ends. The spreadsheets are replaced by the verified central dashboard, and Excel becomes an export format rather than the system of record.
- The Board, finance and audit colleagues and the data team, including those overseas, see one verified version of the figures, at the same time and without waiting for a manual step.
- The caregiver receives structured feedback through the return-for-correction path, which raises the quality of field documentation over time.
- Accountability is clear at every stage: two caregivers signed the record, the administrator verified it, and the audit trail shows both.

9. Device Management: Samsung Knox (Option Under Exploration)

The field tablets used for the trial are Samsung devices, which include Samsung Knox, a built-in enterprise security and device-management framework. The Media & Tech Assistant is currently exploring Knox as one option for safeguarding patient data and keeping the ministry-area tablets dedicated to CareLink. The purpose of the exploration is to establish whether Knox can work efficiently for LOMAMO and secure the entire deployment effectively, before any decision is taken to adopt it. No adoption decision has yet been made, and the capabilities and scenarios below describe what is being evaluated, not what is already in place.

9.1 Capabilities being reviewed

- **Kiosk (Single App) Mode.** Locks the tablet to the CareLink application only, preventing caregivers from browsing to base44 itself, installing other apps, or being distracted by unrelated device functions in the field.
- **Remote enrolment and configuration** (Knox Mobile Enrollment, Knox Manage or Knox Configure). Allows LOMAMO's administrator to set up, update or reconfigure every ministry-area tablet centrally, without needing to physically collect each device.
- **Remote lock and wipe.** Allows a lost or stolen tablet to be locked, or its data erased, remotely, protecting confidential patient information.
- **Network and Wi-Fi policy control.** Can restrict which networks a tablet may connect to, reducing the risk of data being transmitted over unsecured public Wi-Fi.
- **App and settings restrictions.** Can block installation of unauthorised applications and prevent changes to device settings that are not relevant to fieldwork.
- **Usage scheduling and device tracking.** Can restrict device use to expected working hours and provide visibility of device location and status for asset management.

9.2 Use cases the exploration will test

- A caregiver's tablet is lost during a village home visit: the administrator remotely locks the device and, if it cannot be recovered, wipes patient data to prevent a confidentiality breach.
- A new caregiver joins a ministry area: the administrator enrolls their tablet centrally via Knox and it arrives already locked into CareLink, with the correct settings, requiring no manual configuration in the field.
- A caregiver is tempted to use the ministry-area tablet for personal browsing or unrelated apps during working hours: Kiosk Mode prevents this by restricting the device to CareLink only, in line with the single-purpose use intended for the tablet.
- Head office needs to confirm that all ministry-area tablets are running the latest version of CareLink ahead of a reporting deadline: Knox Manage would allow this to be pushed and verified remotely across all devices at once.

9.3 Next steps for the evaluation

The Media & Tech Assistant is to establish which licensing tier would be needed, for example Knox Configure for basic provisioning against Knox Manage or Knox Suite for full mobile-device management, what each tier costs at the scale of ten ministry areas, and how Kiosk Mode and remote wipe behave on the exact tablet model in use. The findings, together with a clear recommendation for or against adoption, should be reported back to management before rollout. If Knox is adopted, it should sit alongside the application-level access model in Section 6 as an added layer of protection, not as a replacement for it.

10. Progressive Web App or Standalone Mobile Application?

The trial team is also asking whether converting CareLink from its current progressive web app (PWA) format into a standalone Android application would be preferable. Both approaches have genuine advantages, summarised below.

CONSIDERATION	CURRENT WEB APP (PWA)	STANDALONE ANDROID APP
Offline reliability	Dependent on browser cache and storage limits.	More robust native local storage and background synchronisation.
Native device features	Limited access to camera, notifications and the file system.	Full access to camera, push notifications and device storage.
Distribution and updates	Instant update via the web link; no installation required.	Requires packaging, installation and a distribution or update channel.
Platform reach	Works across Android, iOS and desktop browsers.	Android only, unless separate builds are made for other platforms.
Development cost and time	Lower; builds on the current platform.	Higher; a longer, separate development effort.

On balance, the recommendation is to stabilise and correct the core web-based workflow first, and to evaluate a standalone application only afterwards, as a later-phase enhancement once field requirements have been fully proven. Converting to a native application before the underlying workflow is settled would risk duplicating development effort on features that may still change.

11. Positive Observations

Alongside the improvements identified above, the trial confirmed a number of features that already work well and provide a sound basis to build on. The introduction of the app (digitization) is also an advantage in various ways:

- The system preserves data entered in a form that is not completed, allowing the caregiver to return later and continue without losing earlier entries.
- Couple registration is achievable in principle, since a patient's spouse can be linked to their record from assessment stage onward.
- Discharge functionality could summarise the full home-visit history for a patient automatically, supporting continuity of care and reducing manual reconstruction of records.
- The proposed login structure, with work email, ministry-area selection, a PIN for area-level access and full oversight for the director and administrators, offers a sound basis for secure, role-based access.
- The administrator's current practice of reviewing and signing every paper form gives LOMAMO a proven quality-control step, which the digital workflow in Section 8 can reproduce and strengthen.
- Automatic calculation of days in the programme is a straightforward but valuable feature once enrolment dates are captured.
- The field tablet (Section 13) is fit for purpose, with battery performance well suited to village ministry areas without mains electricity, and a suitable screen size.

12. Prioritized Recommendations

PRIORITY	RECOMMENDATION	EXPECTED BENEFIT	TIMING
Critical	Fix the Submit button fault that sometimes requires several taps before a completed form registers as submitted.	Prevents lost, delayed or duplicated field records.	Before rollout
Critical	Correct date selection, enable multi-form offline saving and reviewing or editing, resolve blocked-button notifications, and add incomplete-form validation.	Prevents inaccurate, incomplete and delayed field records.	Before rollout
Critical	Apply conditional (skip) logic across Sections 3 to 9, including the reordered medication-default sequence, the new wound-care opening question, and the structured ARV and TB medication lists.	Shorter, cleaner forms with fewer irrelevant or error-prone fields.	Before rollout
Critical	Define data storage, ownership, access rights, security and audit controls, using the questions for the builder in Section 14.	Protects patient information and establishes accountability.	Before rollout
Critical	Publish CareLink with its own independent sign-in layer, separate from the base44 builder account, and implement the master-administrator and caregiver-PIN access model.	Removes an inappropriate developer-level login and gives orderly, accountable, traceable access.	Before rollout

High	Implement the submission, verification and publishing workflow of Section 8: the administrator's console, return-for-correction, digital approval with name and timestamp, and publication to the central dashboard.	Preserves the office's quality control digitally and ends manual entry into Excel.	Before rollout
High	Add two signature blocks to every form, for the lead and the accompanying caregiver, with on-screen signature capture, offline support and a reasoned solo-visit exception (Section 3.9).	Every visit is attributable to the staff who actually made it, as with the signed paper forms.	Before rollout
High	Add the full patient-not-at-home workflow: absence reasons (hospital visit, unknown or other, and at school, elsewhere or hospitalised for minors), bag types (normal, family, couple, special) with Bag 1 or Bag 2 recorded, couple-bag spouse confirmation with the spouse at-home tick, recipient relationship, phone or no-phone status, and explicit no-bag outcomes.	Makes the workflow realistic and improves traceability of support delivered.	Before rollout
High	Introduce patient and ministry-area selection with automatic caregiver details, auto-populated recurring fields (last visit date, contact information) and a relief-caregiver option.	Reduces duplicate entry, speeds up visits and strengthens continuity of care.	Before rollout
High	Track weight measurements so each patient is weighed at least once every three weeks or three consecutive visits, with automatic prompts and overdue flags.	Secures a verifiable nutrition and health monitoring schedule.	Before rollout
High	Add a privacy-preserving proofreading and terminology checker for free-text fields.	Improves grammar and professional presentation without external AI processing or alteration of caregiver facts.	Before rollout
High	Develop the organisation-wide, ministry-area and per-patient management dashboards, including food bags by type and bag number.	Provides timely information for planning, accountability and continuity of care.	Before rollout
High	Conclude the Media & Tech Assistant's evaluation of Samsung Knox (licensing tier, cost, Kiosk Mode, remote lock and wipe) and decide on adoption.	Confirms whether tablets can be locked to CareLink and patient data protected on lost or stolen devices.	Before or early rollout
Medium	Record GPS household locations and dates of birth, with ages calculated automatically.	Stand-in caregivers can locate households, and minor-specific absence options become possible.	Before rollout

Medium	Add seven-day multivitamin supply logic and a separate “taking them properly” adherence indicator.	More accurate, realistic adherence tracking.	Before rollout
Medium	Enable flexible, needs-based home-visit frequency, including multiple weekly visits for active wound-care cases and fortnightly options where appropriate.	Matches the visit schedule to clinical need rather than a single fixed interval.	Before rollout
Medium	Digitise the remaining programme forms (assessment, hospital in-patient and out-patient, discharge, patient deceased and disclaimer) within the patient record, including spouse and couple linkage and a days-in-programme counter.	Completes the digital transformation agenda and supports PDCA preparation.	Phased development
Medium	Evaluate a standalone Android application once the web workflow is stabilised.	May improve offline reliability and native-device features; trade-offs are set out in Section 10.	Later phase

13. Tablet and Field-Device Experience

The trial found the Samsung A11+ 5G tablet adequate for running the Chrome web browser and the CareLink web application across all four ministry areas tested (Section 2.1), and the screen size was suitable for completing the form in the field. A more appropriate book-style protective cover was recommended, which will be sourced through TechCloud Enterprises, the supplier of the first trial tablet unit.

Battery performance was a particular strength. It was found to be very good for field work of this kind, which matters directly for LOMAMO's operations, because home visits in the village ministry areas, including Ombeyi and Manyatta, routinely take place where there may not be mains electricity available for charging during the day.

Caregivers can go out for a full day of village home visits with confidence and without carrying a power bank or other back-up power source, because the device is sufficiently power-efficient to last the working day on a single charge.

These findings indicate that the current tablet hardware can support the pilot and early implementation, provided that the application's field workflow is improved as set out above, and subject to the outcome of the Samsung Knox evaluation in Section 9.

14. Questions for the System Builder

CareLink completely changes the protocol and procedure for handling patient records at the office. It therefore raises administrative, data-storage and management questions that LOMAMO must put to the builder of the system before the platform is adopted. The questions below should be answered first, so that rollout decisions are made on confirmed facts rather than assumptions. Where an answer is unsatisfactory, the item should be raised as a risk and resolved during the implementation phase of the suggested recommendations.

14.1 Administration, data storage, ownership and security

I. Where exactly is patient data physically stored, meaning which provider and which country, and who legally owns that data?

II. Is data encrypted while it travels between the tablet and the server, and while it sits at rest on the server?

III. Can anyone outside LOMAMO, including base44 staff or its sub-processors, technically view patient records? Does base44 process any of this data with artificial-intelligence services?

14.2 Verification, publishing and dashboards

IV. For this data, are LOMAMO Foundation Kenya administrators able to have a console or dashboard to view and actually work with (full permission access) all the data as entered by caregivers in the field?

14.3 Costs, support and sustainability

V. What are the ongoing builder site subscription costs, and how do they change as the number of users, records or ministry areas grows?

VI. Can the application be moved to a LOMAMO-owned domain, for example carelink.lomamo.org, so that the web address itself is under the Foundation's control?

VII. What support and response times does the builder commit to when the system fails in the field, and is this covered by a written agreement?

VIII. If the builder becomes unavailable, is the application's design documented so that another developer can continue it?

15. Conclusion

The CareLink trial demonstrates that LOMAMO Foundation Kenya has a valuable opportunity to modernise its home-visit data-collection process. The platform already provides a useful foundation, and its offline orientation is well suited to field work, including the village ministry areas where internet or cellular connectivity may be problematic, electricity may not be available, and a power-efficient device is genuinely necessary.

The trial, conducted across four of LOMAMO's ten ministry areas (Nyalenda, Ahero, Ombeyi and Manyatta), with three caregivers completing fifteen patient entries, was a proportionate and adequate basis for a pilot of this kind, and has been valuable in identifying practical issues before full implementation. These extend well beyond navigation and offline behaviour into the precise clinical and administrative logic each section of the form should follow, into how the system should be governed and accessed, and into how the field devices themselves should be secured and managed.

The further recommendations consolidated in this revision, covering absence reasons, family and couple bags, Bag 1 and Bag 2 recording, spouse confirmation when a couple bag is issued, the structured ARV list, weight monitoring, GPS household locations, patient ages and the two caregiver signatures on every form, have been incorporated so that the system reflects the programme's actual field protocol rather than an approximation of it.

Two structural changes complete the picture. Every form the programme uses, from assessment to disclaimer, must move into the same digital patient record (Section 3.8), because the manner in

which data is handled has changed. And every submitted form, signed at the point of care by both visiting caregivers (Section 3.9), must pass through the administrator's verification before it is published to the central dashboard (Section 8), preserving the office's quality control and giving the Board, the finance and audit colleagues and the data team one verified version of the truth.

The recommended direction remains to continue developing CareLink, but to judge success by whether it makes the caregiver's data capture and entry work simpler, more accurate and more efficient. The system should minimise unnecessary typing, prevent avoidable data-entry errors, preserve a complete patient history and make relevant management information immediately available.

Just as importantly, CareLink should be published as a proper, self-contained application with its own administrator-controlled sign-in, so that caregivers only ever interact with the finished tool and never with the base44 platform used to build it. The Media & Tech Assistant's exploration of Samsung Knox should be concluded before wider rollout, so that the decision on device management is made on confirmed findings, and the questions in Section 14 should be answered by the builder so that the office's administrative, data-storage and management concerns are settled before the new protocol replaces the old one.

For free-text documentation, LOMAMO should favour a privacy-preserving proofreading and terminology function rather than external AI processing. This would raise the standard of caregiver documentation while ensuring that the caregiver remains the source and final authority for the factual content of the record.

With the critical improvements implemented and a controlled caregiver-acceptance phase completed, CareLink can develop from a digital home-visit form into a coherent patient and programme information system that supports field efficiency, continuity of care, data quality, accountability and evidence-based management.

OVERALL RECOMMENDATION:
PROCEED WITH CARELINK APP REFINEMENT AND CONTROLLED IMPLEMENTATION